Lost in One Index: Measuring How Language and Modality Gaps Compound in English–Indonesian Multimodal Scholarly Search

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Abstract

Scholarly knowledge in Indonesia is split across languages: research papers are published in English, while lectures and theses are produced in Indonesian. Retrieval systems that embed all sources into a single vector index promise to bridge this divide, but their cross-lingual behavior is rarely measured. We build a bilingual multimodal test collection — ten English papers (2,720 text chunks, 67 figures), eight English and eight Indonesian lecture videos (2,101 transcript segments), and 15 translation-paired queries — and measure how retrieval changes when the same question is asked in English versus Indonesian. Under an English-only CLIP encoder, retrieval silos by language: paired queries return nearly disjoint top-10 lists (mean Jaccard overlap 0.019), Indonesian queries reach English papers in only 3 of 15 cases, and the language and modality gaps compound — Indonesian-query-to-figure similarity is the lowest-scoring group in the collection (mean 0.498 vs. 0.812 for same-language transcripts). A multilingual CLIP text encoder closes most of the consistency gap (Jaccard 0.474) but introduces a register bias that floods results with lecture transcripts. Per-group score calibration restores coverage — figures return to up to 7 of 15 queries and every Indonesian query reaches English papers — but halves relevance for Indonesian queries (nDCG@10 0.120 $\rightarrow$ 0.051), exposing a fairness–precision tradeoff. A BM25 baseline outperforms every dense configuration in both languages, aided by code-switched English terminology in Indonesian lectures. We release the collection, relevance judgments, and code.

1 Introduction

A student at an Indonesian university studying deep learning lives between two languages. The primary literature — arXiv papers, conference proceedings — is in English. The explanations that make that literature accessible — university lectures on YouTube, campus theses, national journals — are largely in Bahasa Indonesia. Writing a literature review means searching both worlds, yet search tools handle them separately: an Indonesian question does not find English papers, and an English query does not surface the Indonesian lecture that best explains the concept.

Multimodal retrieval systems that embed text, figures, and lecture transcripts into one shared vector space appear to offer a way out: if all content lives in the same space, one query should retrieve across languages and modalities alike. Prior work on such systems, including our own earlier prototype [1], has documented a *modality gap*: in a shared CLIP index, figures score systematically below text and are almost never retrieved [2]. This paper asks what happens when a *language* dimension is added, using English–Indonesian as the test pair. Indonesian is spoken by more than 270 million people, yet Indonesian-language evaluation resources remain scarce [8, 7], and to our knowledge no prior work measures cross-lingual behavior in multimodal scholarly search.

We make four contributions:

1. **A bilingual multimodal test collection.** Ten English deep-learning papers, eight English and eight Indonesian lecture videos, 15 translation-paired queries (authored and verified by a native speaker), and 999 pooled relevance judgments, all released publicly.

2. **A measurement of the language gap and its interaction with the modality gap.** Under English-only CLIP, translation-paired queries return nearly disjoint results (Jaccard 0.019), and the two gaps stack: Indonesian queries against English figures form the lowest-similarity group in the collection.

3. **An encoder comparison.** A multilingual CLIP text encoder recovers most cross-lingual consistency (Jaccard 0.474) but compresses similarity scores and biases retrieval toward spoken-register content, displacing paper text.

4. **A calibration experiment revealing a fairness–precision tradeoff.** Z-normalizing scores within each (modality, language) group restores coverage across groups but reduces relevance for Indonesian queries by half, showing that balanced exposure and precision are competing objectives in a single shared index.

2 Related Work

Cross-lingual and multilingual retrieval. Cross-language information retrieval has classically relied on translation [3]; multilingual sentence encoders such as LaBSE [4] and knowledge-distilled multilingual CLIP [5] enable direct cross-lingual dense retrieval by aligning languages in one embedding space. Benchmarks such as MIRACL [6] evaluate multilingual text retrieval, and Indonesian NLP resources are expanding through efforts like IndoNLU [7] and NusaCrowd [8]. Multimodal *and* multilingual scholarly retrieval, however, remains unmeasured; our collection targets exactly this intersection.

The modality gap. Liang et al. [2] showed that contrastively trained multimodal encoders place image and text embeddings in systematically different regions of the space. In retrieval over academic content, this manifests as figure starvation: in our earlier system, figures were never retrieved by unified top-k search over a shared CLIP index [1]. The present work measures how this gap interacts with a language gap and evaluates one mitigation.

Multimodal document retrieval. ColPali [9] retrieves document pages as images with vision-language embeddings, and M3DocRAG [10] performs retrieval-augmented generation over multimodal document collections. These systems target visually rich English documents; our setting adds spoken lectures and a second language.

LLM-based relevance judgment. Recent work shows large language models can produce relevance judgments that correlate with human assessors at useful levels [11, 12]. We adopt this methodology, with guidelines, per-decision conventions, and known biases documented alongside the released judgments.

3 Test Collection and System

3.1 Corpus

The corpus covers deep learning broadly, chosen so that Indonesian lecture material exists for every topic:

- **Papers (English):** ten arXiv papers — word2vec [13], VGG [14], ResNet [15], Transformer [16], BERT [17], Mamba [18], Jamba [19], Mamba-360 [20], Mamba-2 [21], and an empirical study of Mamba-based LMs [22] — yielding 2,720 text chunks and 67 embedded raster figures.
- **Lectures:** eight English videos (including 3Blue1Brown, Stanford CS231N/CS224N, and Michigan lectures) and eight Indonesian videos from university and community channels (Machine Learning Indonesia, Telkom University course recordings, and others), yielding 1,146 English and 955 Indonesian transcript segments. Indonesian transcripts are auto-generated captions; transcription noise is part of the measured condition.
- **Queries:** 15 study questions in translation-paired form (English and Indonesian), spanning architectures, training methods, and model comparisons. The Indonesian phrasings were written and then corrected against academic register norms by the author, a native speaker.

Text and transcripts are chunked to at most 50 words to respect the CLIP text encoder's 77-token input limit. Figures carry captions parsed from PDF text or generated by LLaVA-7B [23], and are embedded as the average of image and caption embeddings. All items carry modality and language metadata. The index contains 4,888 vectors of dimension 512, searched exactly with FAISS [24] inner product over unit-normalized vectors.

3.2 Retrieval configurations

We compare, at k=10 over identical corpora:

- **CLIP**: English CLIP ViT-B/32 for text and images.
- **mCLIP**: a multilingual text encoder distilled into the CLIP image space [5], paired with the same image encoder.
- **BM25**: lexical retrieval [25] over every textual representation (chunks, captions, transcripts), both languages mixed.
- **Calibrated variants**: for each dense encoder, similarity scores are z-normalized per query *within each (modality, language) group* before ranking, so that no group is advantaged by the scale of its score distribution.

3.3 Measures

For every configuration and query language we report: the modality and content-language distribution of retrieved items; per-query coverage (queries returning $\geq$1 figure, $\geq$1 item of each content language); *EN$\leftrightarrow$ID consistency* — for each translation-paired query, the Jaccard overlap between the English query's and the Indonesian query's result sets, averaged over pairs (a language-fair system approaches 1); and graded effectiveness (P@10, nDCG@10) against pooled binary relevance judgments. Judgments cover all 999 items retrieved by any configuration and were produced by a large language model following written guidelines (strict standard; term mention $\neq$ relevance; items judged on meaning regardless of language; figures judged by inspecting the image). Guidelines, per-convention notes, and known biases are released with the data; the single-judge design is a limitation (Section 6).

Table 1: Retrieved items by modality and content language (15 queries × top-10). Coverage counts queries returning ≥1 figure / ≥1 English-content item / ≥1 Indonesian-content item.

Config	Query lang.	Text	Fig.	Trans.	EN	ID	Coverage (fig/EN/ID)		
CLIP	EN	74	0	76	148	2	0/15	15/15	2/15
CLIP	ID	11	0	139	16	134	0/15	3/15	15/15
mCLIP	EN	13	0	137	22	128	0/15	8/15	15/15
mCLIP	ID	5	0	145	8	142	0/15	5/15	15/15
BM25	EN	85	4	61	146	4	2/15	15/15	1/15
BM25	ID	18	2	130	25	125	1/15	7/15	15/15

Table 2: Query–item cosine similarity by (modality, content language), mean ± std over 15 queries × all items in the group.

	CLIP		mCLIP	
Group (n)	EN queries	ID queries	EN queries	ID queries
Text EN (2720)	.714 ± .075	.649 ± .072	.835 ± .033	.843 ± .035
Figure EN (67)	.556 ± .043	.498 ± .041	.615 ± .034	.617 ± .033
Transcript EN (1146)	.761 ± .049	.703 ± .050	.860 ± .024	.877 ± .026
Transcript ID (955)	.718 ± .051	.812 ± .056	.880 ± .023	.899 ± .027

4 Results

4.1 English-only CLIP silos retrieval by language

Table 1 shows the modality and language distribution of retrieved items. Under CLIP, English queries retrieve almost exclusively English content (148/150 items) and Indonesian queries almost exclusively Indonesian content (134/150); only 3 of 15 Indonesian queries retrieve any English paper content at all. The consistency between translation-paired queries is near zero (Jaccard 0.019, overlap@10 0.033): *the same question, asked in the other language, returns an almost entirely different result list*. Figures are never retrieved by either query language, replicating the modality gap on a corpus 2.3× larger than in our earlier study [1].

4.2 The gaps compound

Table 2 reports the mean cosine similarity between queries and *all* indexed items, grouped by (modality, content language). Two structures are visible. First, the *language gap*: under CLIP, Indonesian queries score same-language transcripts at 0.812 but English paper text at only 0.649 — a same-language bonus of +0.16 that explains the siloing in Section 4.1. Second, the *modality gap*: figures score far below all text groups for both query languages. The two effects stack: the Indonesian-query→English-figure group (0.498) is the lowest cell in the table, and its *maximum* (0.622) barely reaches the *mean* of the competing same-language transcript group. Content that is both cross-language and cross-modality is doubly disadvantaged.

4.3 Multilingual encoding closes the language gap but opens a register gap

Under mCLIP the per-group means become nearly identical for English and Indonesian queries (Table 2, right), and consistency between paired queries rises from 0.019 to 0.474 Jaccard (overlap@10 0.593) — the encoder substitution alone removes most of the language siloing. But the distributions

Table 3: Effectiveness against pooled relevance judgments (999 items, 127 relevant).

	P@10		nDCG@10	
Configuration	EN	ID	EN	ID
BM25	**.253**	**.180**	**.328**	**.241**
CLIP raw	.193	.113	.238	.120
CLIP calibrated	.193	.047	.259	.051
mCLIP raw	.160	.113	.227	.155
mCLIP calibrated	.160	.067	.218	.092

also compress (standard deviations halve) and reorder: transcripts of *both* languages now outscore paper text, and retrieval floods with lecture segments (137–145 of 150 items; Table 1), displacing the papers that hold the most authoritative content. Figures remain at zero. The language bias is traded for a register bias, and the underlying lesson generalizes: *raw similarity scores are not comparable across content groups, and which group wins is an artifact of the encoder's training distribution.*

4.4 Calibration restores coverage but costs precision

Z-normalizing scores within each (modality, language) group forces the groups into comparable ranges. Coverage responds immediately: figures appear for up to 7 of 15 queries (mCLIP-calibrated: 5/15 for Indonesian queries), every Indonesian query retrieves English paper content (3/15 $\rightarrow$ 15/15 under CLIP), and the transcript flood recedes (mCLIP, ID queries: 145 $\rightarrow$ 48 transcript items). Paired-query consistency also peaks (Jaccard 0.505, overlap@10 0.640, mCLIP-calibrated).

Relevance tells the other half of the story (Table 3). Calibration leaves English-query effectiveness roughly unchanged (CLIP nDCG@10 .238 $\rightarrow$ .259; mCLIP .227 $\rightarrow$ .218) but *halves* it for Indonesian queries (CLIP .120 $\rightarrow$ .051; mCLIP .155 $\rightarrow$ .092). Forcing starved groups into the top-10 promotes items that are on-group but off-topic: with only 67 figures and a strict relevance standard, a guaranteed figure slot is usually filled by an irrelevant figure. Balanced exposure and precision are competing objectives; a single shared index satisfies at most one at a time.

4.5 BM25 outperforms all dense configurations

BM25 achieves the best effectiveness in both query languages (Table 3) while retrieving *some* cross-modal content (via figure captions) and even some cross-lingual content: 7 of 15 Indonesian queries reach English material lexically. The bridge is code-switching — Indonesian technical lectures routinely embed English terms (*attention*, *transformer*, *layer*), giving lexical matching shared vocabulary across the language boundary. On a domain-specific corpus of this size, dense retrieval's semantic generalization does not yet pay for its calibration problems.

5 Discussion

Our measurements support three practical conclusions. First, *single-index dense retrieval is unsafe across languages and modalities without calibration*: which content gets retrieved is governed by encoder-induced score offsets (language, register, modality), not only by relevance. Second, *each mitigation trades one bias for another*: multilingual encoding removes the language offset but introduces a register offset; per-group calibration equalizes exposure but promotes irrelevant items from sparse groups. Systems that need both fairness and precision should retrieve per group and merge with a relevance-aware policy — learned fusion, or lexical–dense hybrids that exploit code-switching — rather than rank raw or naively calibrated scores in one list. Third, for low-resource-language deployments, *BM25 remains a strong default*, and evaluations of multilingual dense retrievers should always include it.

6 Limitations

The collection is small (one topic domain, 4,888 items, 15 query pairs) and covers one language pair; effect sizes are large and consistent, but generalization to other languages and domains is untested. Relevance was judged by a single LLM assessor; although the methodology follows recent practice [11, 12] and all guidelines, conventions, and judgments are released for audit, human verification — even of a sample — would strengthen the conclusions. Indonesian transcripts are auto-generated captions, so ASR noise and caption quality are confounded with language effects (we treat this as part of the realistic condition, since it is what any deployed system would index). Finally, snippet truncation during judging may undercount the relevance of long text chunks relative to transcripts.

7 Conclusion

We presented the first measurement, to our knowledge, of how language and modality gaps interact in multimodal scholarly retrieval, using a new English–Indonesian test collection. An English-only encoder silos retrieval by language; a multilingual encoder fixes alignment but not score comparability; calibration fixes exposure but not relevance; and a lexical baseline quietly outperforms them all. The collection, judgments, and code are released to support work on the missing piece: retrieval that is simultaneously language-fair, modality-fair, and precise.

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